

Abolfazl Arab (Abe)

Title	Research topic	Contact
PhD Student at Shokat Lab, UCSF	Computational Chemical Genomics	abolfazl.arab@ucsf.edu

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Education

San Francisco	University of California, San Francisco (UCSF)
Sep 2024 – current	UCSF – Graduate Program in Pharmaceutical Sciences and Pharmacogenomics (PSPG)
Sep 2025 – current	UCSF / UC Berkeley – Designated Emphasis in Computational Precision Health (CPH)
Tehran, Iran	University of Tehran
Sep 2013 – July 2018	B.Sc. Cellular and Molecular Biology, School of Biology, College of Science
Sep 2018 – Sep 2021	M.Sc. Nano-biomimetics, Department of Life Science Engineering

Publications

- O’Loughlin TA*, Arab A*, ..., Forment JV, Gilbert LA, **A chemical-genetic interaction map of the DNA damage response reveals an MRGBP-PAX7-IREB2-PRDX1 axis regulating iron-induced DNA damage**, [Manuscript under review at *Nature Chemical Biology*]
 - Ge A*, Arab A*, Dai R*, ..., Goodarzi H, Gilbert LA, **A multiomics approach reveals RNA dynamics promote cellular sensitivity to DNA hypomethylation**, *Scientific Reports* (2024) | doi: [10.1038/s41598-024-77314-9](https://doi.org/10.1038/s41598-024-77314-9)
- * equal contribution first authors

Research and Work Experiences

San Francisco	Graduate Student University of California, San Francisco (UCSF)
Sep 2024 – current	Title: Graduate Student – Supervisor: Dr. Kevan Shokat Projects: Apply Computational Chemical Genomics approaches for cancer therapeutic design - Study the p53 tumor suppressor reactivation and targeting epi-genetic regulators - Chemical synthesis of novel compounds to exploit MDM2 accumulation as a therapeutic vulnerability
Fall 2024	Title: Rotation Student – Supervisor: Dr. Katie Pollard - Initiate the Functional Genomics Interpreter (FunGI) as an open-source platform - Build a framework to evaluate sequence-to-function model predictions on genetic variants
Spring 2025	Title: Rotation Student – Supervisor: Dr. Nichole DelRosso Identify synergistic activation domains using HT-recruit to measure protein domain’s effect on transcription
Palo Alto, USA	Gilbert Lab and Multi-Omics Tech Center Arc Institute
Jul 2022 – Sep 2024	Title: Research Associate II – Supervisor: Dr. Luke Gilbert Research: Developing and applying dCas9 and dCas12a CRISPRi/a as functional genomic tools for drug discovery - Developed a flexible and generalizable analysis platform for high-content CRISPR screening – ScreenPro2 - Built graph representation of gene-gene and drug-gene interactions to study DNA damage response - Studied epi-{genome,transcriptome} crosstalk in Acute Myeloid Leukemia (AML) - Designed kinome targeting Cas12a library to study synergistic dependencies in higher-order perturbation - Evaluated CRISPROff specificity on target gene’s expression and DNA methylation in primary human T Cells
Nov 2023 – Sep 2024	Title: Senior Research Associate – Supervisor: Dr. Nianzhen Li Performed bulk and single cell multi-omics data analysis and developed pipelines and GUIs
Boston, USA	Summer Institute in Biomedical Informatics (SIBMI) DMBI, Harvard Medical School
Jul 2019 – Sep 2019	Title: Post-bac Fellow – Mentor: Dr. Hadi Amiri, Supervisor: Dr. Isaac S. Kohane Research: Biological similarities among rare disease patients at Undiagnosed Diseases Network (UDN) - Clustered patients with undiagnosed disease using textual information in referral letters and EHR data - Built automated patient phenotyping workflow using NLP representation learning techniques
Apr 2023 – Apr 2024 (Remote)	Title: Volunteer Researcher at Zitnik Lab – Supervisor: Dr. Marinka Zitnik Project: Defining new generalizable data loaders and tasks in artificial intelligence research on therapeutics - Wrote new data loader module for knowledge graphs in Therapeutics Data Commons (TDC) - Extended Precision Medicine Knowledge Graph (PrimeKG) by including OMIM gene and phenotypes entries
San Francisco, USA	Goodarzi and Gilbert Labs UCSF
Sep 2021 – Jul 2022	Title: Scientific Research Assistant II – Supervisor: Dr. Luke Gilbert
Sep 2019 – Jul 2020	Title: Junior Specialist II – Supervisor: Dr. Hani Goodarzi - Performed bulk and single cell RNA-seq data analysis of in-vitro or in-vivo drug and gene perturbation studies - Implemented analytic workflow for RNA expression, stability, splicing, and m⁶A methylation analysis - Collaborative NGS data analysis with laboratories in UCSF, Rockefeller, Yale, Harvard, etc.
Boston, USA	Cancer Cell Line Encyclopedia (CCLE) Project Broad Institute of Harvard and MIT
Aug 2018 – Oct 2018	Title: Volunteer Researcher – Adviser: Dr. Mahmoud Ghandi Integrated and assessed co-expression patterns of microRNAs and target genes across cancer cell lines
Tehran, Iran	Stem Cell Technology Research Center, Ltd. & University of Tehran
Feb 2017 – Aug 2018	Title: Volunteer Researcher – Supervisor: Dr. Ehsan Arefian Studied microRNA regulation of target gene expression in Glioblastoma cancer cells and patients

Honors and Awards

- Award winner, **JTX + CSHL Biological Data Science 2024 Scholarship**, The James P. Taylor Foundation – Fall 2024
- Funded Post-bac Fellow, **Summer Institute in Biomedical Informatics (SIBMI)**, **Harvard Medical School** – 2019.
- 2nd team at **Startup Weekend**, 2nd Young Scientists Festival, University of Tehran – 2018.
- Awarded student journal of science, Iranian **National Student Journals Festival** – 2016; 2017.
 - DeNA issue 14 – Roles: Editor in chief and layout design
 - DeNA issue 10 – Roles: Secretary, interviewer, and layout design
- Awarded for the **Best Innovative Idea**, Domestic Festival at University of Tehran – 2016.
 - Title: “A development roadmap of scientific student associations”
- Honored to join **high School for the Gifted**, National Organization for Development of Exceptional Talents (**Sampad**) – 2009.

Software and tools

Developer and maintainer:

- **ezformer** Run enformer like models, get prediction tracks, and perform benchmarking – <https://github.com/abearab/ezformer>
- **ScreenPro2** Flexible CRISPR screen analysis for large scale oligo libraries – <https://github.com/ArcInstitute/ScreenPro2>
- **FunGI** Functional Genomics Screening (e.g. CRISPR Screens) Datasets, Tools, and Resources – <https://github.com/FunGI-cap>
- **pager** Pathway enrichment analysis, PAGE algorithm run and downstream analysis – <https://github.com/abearab/pager>

Contributor:

- **TDC** Therapeutics Data Commons – <https://tdcommons.ai> | <https://github.com/mims-harvard/TDC>
- **CanDI** A global cancer data integrator – <https://github.com/GilbertLabUCSF/CanDI>
- **PrimeKG** Precision Medicine Knowledge Graph – <https://github.com/mims-harvard/PrimeKG>

Contributed in > 20 more **open-source projects** in GitHub repositories.

Presentations

Conferences:

- Arab A, *et al.*, **A path toward development of FunGI – Functional Genomics Interpreter**, Poster Presentation, *Biological Data Science Conference, Cold Spring Harbor Laboratory* (2024)
- Arab A, Gilbert L, **Integrative systems biology approach reveals RNA dynamics that promote cellular sensitivity to DNA hypomethylation as potential to design new combination therapy strategies**, Poster Presentation, *Bay Area RNA Club 2023*, UCSF, San Francisco (2023)
- Arab A, Gilbert L, **A multiomics approach identifies genes potentially regulated through RNA m6A methylation and promote cellular sensitivity to DNA hypomethylation**, Poster Presentation, *1st NYC RNA Symposium 2023*, The Rockefeller University, NYC (2023)
- Arab A, *et al.*, **Demonstration of gene regulatory network in DNA repair phenotype and gene expression at single cell resolution**, Poster Presentation, *1st International and 10th National Iranian Conference on Bioinformatics*, Kish, Iran, (2022)
- Arab A, *et al.*, **Predicting an application of miR-190b: Apoptotic enhancement effects on glioblastoma multiform**, Poster Presentation, *the 3rd Iranian Conference on Systems Biology*, Tehran, Iran, (2018)

Co-author Publications

- Goudy L, ..., Arab A, *et al.*, **Integrated Epigenetic and Genetic Programming of Primary Human T Cells**. [Manuscript under review at *Nature Biotechnology*]
- Hsiung CC, ..., Arab A, *et al.*, **Higher-order combinatorial chromatin perturbations by engineered CRISPR-Cas12a for functional genomics**, *Nature Biotechnology* (2024) | doi: [10.1101/2023.09.18.558350](https://doi.org/10.1101/2023.09.18.558350)
- Arshadi AK, ..., Arab A, *et al.*, **Functional microRNA-Targeting Drug Discovery by Graph-Based Deep Learning**, *Patterns* (2024) | doi: [10.1016/j.patter.2023.100909](https://doi.org/10.1016/j.patter.2023.100909)
- Richter M, ..., Arab A, *et al.*, **Inhibition of muscarinic receptor signaling protects human enteric inhibitory neurons against platinum chemotherapy toxicity**, *bioRxiv* (2023) | doi: [10.1101/2023.03.08.531806](https://doi.org/10.1101/2023.03.08.531806)
- Majd H, ..., Arab A, *et al.*, **Deriving Schwann cells from hPSCs enables disease modeling and drug discovery for diabetic peripheral neuropathy**, *Cell Stem Cell* (2023) | doi: [10.1101/2022.08.16.504209](https://doi.org/10.1101/2022.08.16.504209)
- Xiao J, Sharma U, Arab A, *et al.*, **Propagated circulating tumor cells uncovers the role of NFκB and COP1 in metastasis**, *Cancers* (2023) | doi: [10.3390/cancers15061831](https://doi.org/10.3390/cancers15061831)
- Furuya H, ..., Arab A, *et al.*, **PAI-1 is a potential transcriptional silencer that supports bladder cancer cell activity**, *Nature Scientific Reports* (2022) | doi: [10.1038/s41598-022-16518-3](https://doi.org/10.1038/s41598-022-16518-3)
- Yogodzinski C, Arab A, *et al.*, **A global cancer data integrator reveals principles of synthetic lethality, sex disparity and immunotherapy**, *Genome Medicine* (2021) | doi: [10.1186/s13073-021-00987-8](https://doi.org/10.1186/s13073-021-00987-8)
- Zhu X, ..., Arab A, *et al.*, **Functional Genomics In Vivo Reveal Metabolic Dependencies of Pancreatic Cancer Cells**, *Cell Metabolism* (2020) | doi: [10.1016/j.cmet.2020.10.017](https://doi.org/10.1016/j.cmet.2020.10.017)
- Tabibkhoei A, ..., Arab A, *et al.*, **Profiling of novel circulating microRNAs as a non-invasive biomarker in diagnosis and follow-up of high and low-grade gliomas**, *Clinical Neurology and Neurosurgery* (2020) | doi: [10.1016/j.clineuro.2019.105652](https://doi.org/10.1016/j.clineuro.2019.105652)

Mentorship and Teaching Experiences

- Teaching Assistant for “**Foundation I**” course for **PharmD Degree Program**, School of Pharmacy, UCSF – Summer 2025
- Mentored **rotation student** Meri Okorie, **Analysis of pooled RNA-seq dataset for drug screening**, Goodarzi Lab, UCSF – Winter 2022
- Mentored **high school student** Sonakshi Bhalla, **Implement a WES analytic pipeline**, Goodarzi Lab, UCSF – Winter 2022
- Content designer and lecturer, “**Introduction to RNA-seq data analysis**”, Goodarzi and Gilbert Labs, UCSF – Summer 2022
- Teaching Assistant for **Molecular Genetics**, University of Tehran – Spring 2019
- Teaching Assistant for **Mathematics I**, University of Tehran – Fall 2014

Scientific Skills

- **Biology, Bioinformatics, and Biomedical Informatics:**
 - **Machine Learning in Genomics:** perform prediction and fine-tune sequence-to-X models, alphagenome, enformer, akita.
 - **Functional Genomics:** CRISPR{i, a, off}; Cas{9, 12, 13}; large scale CRISPR screens; guide-RNA library design and cloning.
 - **Molecular and Cellular Biology:** Cloning, PCR, fluorescence assay; Cell culture, lentivirus vectors, flow cytometry, cell sorting.
 - **Systems and Synthetic Biology:** Fusion protein engineering, epigenetic editing, gene regulatory networks, multi-omics.
 - **Bulk and Single-Cell Multi-Omics:** scRNA-seq, CITE-seq, perturb-seq; Preprocessing, QC, multi-modal data integration.
 - **Next Generation Sequencing:** mRNA-seq, meRIP-Seq, Ribo-Seq, CLIP-seq; WES, WGBS-seq, RRBS-seq, XRBS, ChIP-Seq.
- **Math, statistics, and computer science:**
 - **Concepts:** Mathematical thinking, algorithms, statistical inference, graph theory, deep leaning.
 - **Machine Learning:** NLP, clustering, representation learning. PyTorch, TensorFlow, Auto-encoder.
 - **Languages and Environments:** Python, R, bash, awk, MATLAB, Rust. VS-Code, RStudio, PyCharm, Jupyter.
 - **Package and Pipeline Development:** GitHub, Conda, Docker, PyPI, Bioconductor, Nextflow.

Online Courses

- Evaluation of Undiagnosed and Rare Conditions – Offered by **Undiagnosed Diseases Network (UDN)**
- Introduction to **Mathematical Thinking** – Offered by Stanford University at Coursera
- **Artificial Intelligence** for Everyone – Offered by DeepLearning.AI at Coursera
- **Deep Learning** Specialization (x5 courses) – Offered by DeepLearning.AI at Coursera
- **Natural Language Processing** in TensorFlow – Offered by DeepLearning.AI at Coursera
- **Python Programming** tracks and individual courses (>40 hours) – Offered by DataCamp
- **R Programming** tracks and individual courses (>40 hours) – Offered by DataCamp
- Introduction to **Programming with MATLAB** – Offered by Vanderbilt University at Coursera
- How to **Build a Startup** – Offered by The Lean LaunchPad at Udacity

Volunteer Activities and Hobbies

- Secretary, Editor in Chief, or author at DeNA, Student Journal of Biology, University of Tehran – June 2014 - June 2020
 - **Arab A**, Pursue problem-solving biology: What I’ve learned at Harvard, *DeNA* 22 (2020)
 - **Arab A**, Interview with Dr. Mohsen Khosravi M.D., Ph.D., “The Inception of being a Researcher”, *DeNA* 15 (2016)
 - Jadidi M, **Arab A**, CRISPR Gene Therapy System: Novel Tools to Treat Complex Diseases, *DeNA* 15, (2016)
 - **Arab A**, Interview with Dr. Hani Goodarzi Ph.D., “Today’s Scientists Social Status”, *DeNA* 10 (2014)
- Member of Student Association of Biology, University of Tehran – June 2014 - June 2017
- Basketball – player at high school and university college teams, Kong-Fu Toa, Mountain Bike